

PROJECT REPORT

AI-Powered Resume Generator and Analyzer

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

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CANDIDATE'S DECLARATION

I/We hereby certify that the work which is being presented in the project progress report entitled **"AI-Powered Resume Generator and Analyzer"** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering in the Department of Computer Science and Engineering of the Graphic Era Hill University, Dehradun shall be carried out by the undersigned under the supervision of **Mr. Deepak Upadhyay, Assistant Professor**, Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun.

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ABSTRACT

In today's highly competitive job market, a well-structured resume plays a critical role in securing employment opportunities. However, many job seekers lack the knowledge of industry-standard resume formats, ATS (Applicant Tracking System) optimization, and effective presentation of their skills. Simultaneously, recruiters spend significant time manually screening large volumes of resumes, leading to inefficiencies in the hiring process.

This project proposes and implements an AI-Powered Resume Generator and Analyzer, a full-stack web application that addresses both sides of this problem. The system consists of two primary modules: (1) an intelligent Resume Generator that allows users to create professional, ATS-friendly resumes using customizable templates with live preview, and (2) an AI-driven Resume Analyzer that evaluates uploaded resumes against target job descriptions, providing detailed scoring, feedback, and job role recommendations.

The application is developed using the MERN stack (MongoDB, Express.js, React.js, Node.js), with Google Gemini AI as the artificial intelligence backbone for resume optimization, scoring, and feedback generation. Additional features include user authentication, profile image upload with background removal via ImageKit, PDF/DOCX export, live shareable resume links, and resume management capabilities.

The frontend is built with React.js and Tailwind CSS for a modern, responsive user interface, while the backend uses Node.js and Express.js with Multer for PDF file handling. Resume data is stored and managed in MongoDB. The application is deployed on Vercel for scalable cloud hosting.

The system demonstrates the effective integration of artificial intelligence with modern web development practices to automate and enhance the resume building and evaluation process, providing significant value to both job seekers and recruiters.

Keywords: *Resume Generator, Resume Analyzer, MERN Stack, Google Gemini AI, Artificial Intelligence, Natural Language Processing, ATS Optimization, React.js, Node.js, MongoDB, Full-Stack Web Application, ImageKit.*

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ATS	Applicant Tracking System
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DB	Database
DFD	Data Flow Diagram
ER	Entity Relationship
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token

ABBREVIATIONS

Abbreviation	Full Form
MERN	MongoDB, Express.js, React.js, Node.js
ML	Machine Learning
NLP	Natural Language Processing
NPM	Node Package Manager
PDF	Portable Document Format
REST	Representational State Transfer
SRS	Software Requirements Specification
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

NOTATIONS

Notation	Meaning
S	Overall resume score (0–100)
W_i	Weight assigned to section i of the resume
F_i	Feature score for section i
n	Total number of resume sections evaluated
JD	Job Description text input
R	Resume document text (extracted)
$\text{Sim}(R, JD)$	Cosine similarity between resume and job description vectors
T	Token sequence in NLP processing
$P(w \text{context})$	Probability of word w given surrounding context (LLM)
α	Learning rate / threshold for AI scoring
σ	Standard deviation of score distribution
μ	Mean score of resume batch
k	Number of recommended job roles returned
t_{resp}	API response time in milliseconds

CHAPTER 1

INTRODUCTION AND MOTIVATION

1.1. 1.1 Introduction

The rapid advancement of technology has transformed the way organizations recruit employees and how individuals search for employment. In the modern digital era, resumes serve as the primary medium through which job seekers present their qualifications, skills, and experience to potential employers. A well-crafted resume is no longer merely a document — it is a strategic marketing tool that must be carefully tailored to stand out in an increasingly automated recruitment pipeline.

Most modern organizations employ Applicant Tracking Systems (ATS) to automate the initial screening of resumes. These systems parse and score resumes based on keyword matching, formatting standards, and relevance to the job description. Studies indicate that over 75% of resumes are rejected by ATS before they ever reach a human recruiter. This creates a pressing need for intelligent tools that help job seekers build ATS-optimized resumes and understand how their existing resumes are being evaluated.

Simultaneously, the proliferation of Artificial Intelligence (AI) and Natural Language Processing (NLP) has opened new avenues for automating complex tasks that traditionally required human expertise. AI-powered systems can now analyze text documents, extract meaningful patterns, generate coherent content, and provide actionable recommendations — capabilities that are directly applicable to resume building and evaluation.

The "AI-Powered Resume Generator and Analyzer" project addresses these needs by providing an integrated, full-stack web platform that combines modern frontend and backend development with AI capabilities. The application serves two primary user groups: job seekers who wish to create professional resumes, and individuals who wish to evaluate and optimize their existing resumes against specific job requirements.

1.2. 1.2 Motivation

The motivation for this project stems from several real-world challenges observed in the recruitment and job-seeking ecosystem:

- **High ATS Rejection Rate:** Research by Harvard Business School (2021) revealed that 88% of employers report that qualified candidates are rejected by ATS systems due to non-standard formatting or missing keywords. This highlights the urgent need for AI-driven resume optimization tools.
- **Lack of Personalized Feedback:** Most job seekers do not receive any feedback on why their resumes were rejected. An intelligent analyzer can fill this gap by providing actionable, personalized suggestions.
- **Time-Consuming Manual Process:** Creating a professional resume from scratch requires significant time and design expertise. An AI-assisted generator with templates and live preview can dramatically reduce this effort.
- **Growing AI Capabilities:** The availability of powerful language models such as Google Gemini makes it feasible to build sophisticated text analysis and generation pipelines as part of a web application at reasonable cost.
- **Market Opportunity:** The global online recruitment market was valued at USD 28.68 billion in 2023 and is projected to reach USD 56.64 billion by 2030, demonstrating significant demand for tools in this space.
- **Educational Value:** Building this system requires integration of diverse technologies including full-stack web development, cloud services, file processing, and AI APIs — making it an excellent learning and demonstration project.
- **Career **Transition Support**:** A large portion of the workforce seeks career transitions across industries post-COVID. Traditional resumes fail to bridge skill gaps effectively, motivating the need for AI tools that can reframe and reposition existing experience for new target roles.
- **Bias **Reduction in Self-Assessment**:** Job seekers often struggle to objectively evaluate their own resumes due to cognitive bias. An AI-powered analyzer provides neutral, data-driven feedback that human self-review cannot replicate, motivating adoption of such tools.
- **Rise of Remote & Global Hiring:** With remote work normalization, candidates now compete globally rather than locally. This exponentially increases competition and raises the bar for resume quality, creating strong motivation for intelligent resume optimization tools.

1.3. 1.3 Problem Overview

The core problem addressed by this project can be described at two levels:

From the job seeker's perspective: Many qualified candidates fail to present their skills effectively due to poor resume structure, formatting issues, lack of relevant keywords, and unfamiliarity with industry expectations. Without professional resume writing services (which are expensive) or expert guidance, these candidates struggle to optimize their applications.

From the recruiter's perspective: Organizations receive hundreds to thousands of applications per job opening. Manual screening is inefficient, expensive, and prone to bias. Automated tools that can objectively score and rank resumes against job descriptions offer significant value.

Existing solutions are either too expensive (professional resume writing services), too generic (simple template editors), or too isolated (standalone analyzers without generation capability). This project aims to provide a comprehensive, integrated, and intelligent solution accessible to all users through a web browser.

1.4. 1.4 Scope of the Project

The scope of this project encompasses the following areas:

- Development of a full-stack web application using the MERN (MongoDB, Express.js, React.js, Node.js) technology stack
- Implementation of a Resume Generator module with multiple professional templates, live preview, AI-assisted optimization, and export capabilities (PDF and DOCX formats)
- Implementation of a Resume Analyzer module that accepts PDF resume uploads, processes them using AI, and provides comprehensive scoring and feedback
- Integration of Google Gemini AI for natural language processing, resume content optimization, scoring, and job role recommendation
- User authentication and authorization using JWT tokens
- Profile image management with background removal using ImageKit cloud service
- Cloud deployment using Vercel for both frontend and backend services
- The scope includes support for diverse user profiles — freshers, experienced professionals, and career switchers — each requiring different resume structures, section emphasis, and keyword strategies that the system accommodates through flexible templates and AI guidance.
- The project scope encompasses implementation of data security measures including bcrypt password hashing, JWT-based session management, HTTPS-enforced communication, and MongoDB Atlas role-based access controls to protect sensitive user resume data.
- The scope includes the ability for users to maintain and manage multiple versions of their resumes simultaneously — each tailored to a different job role or industry — with individual edit, delete, and share capabilities per resume document.
- The scope covers ensuring the application is fully responsive and functional across major browsers (Chrome, Firefox, Safari, Edge) and across device types including desktops, laptops, tablets, and mobile phones, without requiring any software installation.
- Responsive UI design using React.js and Tailwind CSS

The project does not include direct integration with job portals, employer-side systems, or payment processing. It is designed as a standalone web application for educational and portfolio demonstration purposes.

CHAPTER 2

OBJECTIVES AND PROBLEM STATEMENT

1.5. 2.1 Problem Statement

In the contemporary employment landscape, the process of creating an effective resume and ensuring its relevance for a specific job role has become increasingly complex. The following specific problems have been identified:

2.1.1 Resume Creation Challenges

- Inexperienced job seekers lack the knowledge to structure a professional, ATS-friendly resume
- Designing a visually appealing and well-formatted resume requires design skills that many individuals do not possess
- Managing multiple versions of resumes for different job applications is cumbersome without proper tooling
- Incorporating the right keywords and section structure for specific industries or roles is non-trivial

2.1.2 Resume Evaluation Challenges

- Job seekers do not know how ATS systems evaluate their resumes, leading to preventable rejections
- There is no accessible, affordable tool that provides detailed, section-wise feedback on resume quality
- Understanding which job roles best match one's current skill set requires expert guidance that is not readily available
- Resume optimization is an iterative process without a clear metric for improvement

1.6. 2.2 Objectives of the Project

The primary and secondary objectives of this project are defined as follows:

2.2.1 Primary Objectives

1. To design and implement a full-stack web application that enables users to create, customize, and download professional resumes using AI-assisted templates and live preview functionality.
2. To develop an AI-powered resume analysis engine that accepts PDF resumes, extracts their content, and evaluates them against user-specified job roles or descriptions.
3. To integrate Google Gemini AI to provide intelligent resume optimization suggestions, section-wise scoring, and personalized improvement recommendations.
4. To implement a comprehensive user management system including authentication, profile management, and resume history.
5. To deploy the application on a cloud platform ensuring accessibility, scalability, and reliability.

2.2.2 Secondary Objectives

6. To implement multiple resume templates to accommodate different industries and professional levels.
7. To provide live shareable resume links, enabling users to share their resumes online without requiring a download.
8. To integrate ImageKit for profile photo management, including AI-powered background removal.
9. To implement export functionality supporting both PDF and DOCX formats.
10. To ensure the application is responsive and accessible across different devices and screen sizes.

1.7. 2.3 Expected Outcomes

Outcome	Description	Measurable Metric
Working Resume Generator	Users can create and download resumes	End-to-end creation in < 10 min
AI Resume Optimization	Gemini suggests improvements to content	Relevance score improvement $\geq 20\%$
Resume Analyzer with Score	Uploaded PDF gets scored 0–100	Score accuracy validated against HR experts
Job Recommendations	AI suggests suitable job roles	≥ 5 relevant roles per analysis
Multi-format Export	PDF and DOCX downloads work correctly	Zero formatting errors in exports
User Authentication	Secure login and data management	JWT-based auth, bcrypt password hashing
Cloud Deployment	App live on Vercel	Uptime $\geq 99\%$, page load < 3s

CHAPTER 3

SOFTWARE AND HARDWARE REQUIREMENTS

1.8. 3.1 Hardware Requirements

The following hardware specifications are required for development, testing, and deployment of the AI-Powered Resume Generator and Analyzer system:

3.1.1 Development Machine Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 (8th Gen) / AMD Ryzen 5	Intel Core i7 (11th Gen) / AMD Ryzen 7
RAM	8 GB DDR4	16 GB DDR4 or higher
Storage	256 GB SSD	512 GB NVMe SSD
Display	1366 x 768 px	1920 x 1080 px (Full HD)
Network	10 Mbps Broadband	50+ Mbps Broadband
GPU	Integrated Graphics	Discrete GPU (optional, for rendering)
OS	Windows 10 / Ubuntu 20.04 / macOS 11	Windows 11 / Ubuntu 22.04 / macOS 13+

3.1.2 Server / Deployment Requirements (Vercel Cloud)

Since the application is deployed on Vercel (a cloud-based platform), dedicated physical server hardware is not required. Vercel provides serverless functions with the following characteristics:

- Automatic scaling based on traffic load
- Global CDN (Content Delivery Network) for static assets
- Edge runtime support for low-latency API responses
- MongoDB Atlas cloud database (no on-premise server needed)

1.9. 3.2 Software Requirements

Category	Technology / Tool	Version	Purpose
Runtime	Node.js	v18.x LTS	JavaScript server-side runtime
Package Manager	npm	v9.x	Dependency management
Frontend Framework	React.js	v18.x	UI component library
CSS Framework	Tailwind CSS	v3.x	Utility-first styling
Backend Framework	Express.js	v4.x	RESTful API server
Database	MongoDB	v6.x	NoSQL document database
ODM	Mongoose	v7.x	MongoDB object modeling
AI API	Google Gemini API	gemini-pro	AI text generation & analysis
File Upload	Multer	v1.4.x	PDF/image upload middleware
PDF Processing	pdf-parse	v1.1.x	Extract text from PDF files
Authentication	JSON Web Token (JWT)	v9.x	User authentication tokens

Category	Technology / Tool	Version	Purpose
Password Hashing	bcryptjs	v2.4.x	Secure password storage
Image Service	ImageKit	SDK v4.x	Image upload + BG removal
PDF Generation	html2pdf.js / jsPDF	v1.5.x	Export resume as PDF
DOCX Generation	docx.js	v8.x	Export resume as Word doc
HTTP Client	Axios	v1.x	API calls from frontend
CORS	cors (npm)	v2.8.x	Cross-origin resource sharing
Code Editor	VS Code	v1.85+	Development IDE
API Testing	Postman	v10.x	REST API testing
Version Control	Git + GitHub	v2.x	Source code management
Deployment	Vercel	Latest	Frontend & backend hosting
DB Hosting	MongoDB Atlas	M0 Free Tier	Cloud database hosting
Browser	Google Chrome	v120+	Testing & development

1.10. 3.3 System Requirements Specification (SRS)

The Software Requirements Specification defines the complete behavior and constraints of the system. It serves as a contract between the development team and stakeholders, ensuring all requirements are clearly documented before development begins.

3.3.1 System Overview

The AI-Powered Resume Generator and Analyzer is a client-server web application. The client (React.js SPA) communicates with the backend (Node.js/Express.js REST API) over HTTPS. The backend interfaces with MongoDB Atlas for data persistence, Google Gemini API for AI processing, and ImageKit for media management. PDF files uploaded by users are processed server-side using Multer and pdf-parse.

1.11. 3.4 Functional Requirements

3.4.1 User Management Module

FR ID	Requirement	Priority
FR-UM-01	User shall be able to register using email and password	High
FR-UM-02	System shall validate email format and enforce password strength	High
FR-UM-03	User shall be able to log in using registered credentials	High
FR-UM-04	System shall issue JWT token upon successful login (24h expiry)	High
FR-UM-05	User shall be able to upload and manage profile photo	Medium
FR-UM-06	System shall support AI background removal for profile images	Medium
FR-UM-07	User shall be able to update profile information	Medium
FR-UM-08	User shall be able to log out and invalidate session	High

3.4.2 Resume Generator Module

FR ID	Requirement	Priority
FR-RG-01	User shall be able to create a new resume from scratch	High
FR-RG-02	System shall provide at least 5 professional resume templates	High
FR-RG-03	System shall display a real-time live preview of the resume	High
FR-RG-04	User shall be able to add, edit, and delete resume sections	High
FR-RG-05	System shall allow AI-powered content optimization via Gemini	High
FR-RG-06	User shall be able to export resume in PDF format	High
FR-RG-07	User shall be able to export resume in DOCX format	High
FR-RG-08	System shall generate a unique shareable link for each resume	Medium
FR-RG-09	User shall be able to manage (view, edit, delete) saved resumes	High
FR-RG-10	System shall save resume drafts automatically every 30 seconds	Medium

3.4.3 Resume Analyzer Module

FR ID	Requirement	Priority
FR-RA-01	User shall be able to upload a resume in PDF format	High
FR-RA-02	System shall extract text content from uploaded PDF	High
FR-RA-03	User shall be able to specify a target job title or paste JD	High
FR-RA-04	System shall analyze resume and generate an overall score (0–100)	High
FR-RA-05	System shall provide section-wise scores (Skills, Experience, Education, etc.)	High
FR-RA-06	System shall identify strengths and weaknesses in the resume	High
FR-RA-07	System shall suggest specific improvements for each weak section	High
FR-RA-08	System shall recommend relevant job roles based on resume content	Medium
FR-RA-09	System shall highlight missing keywords compared to target JD	High
FR-RA-10	User shall be able to save and view past analysis reports	Medium

1.12. 3.5 Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Page load time shall be < 3 seconds on a 10 Mbps connection
NFR-02	Performance	AI analysis response time shall be < 15 seconds
NFR-03	Scalability	System shall support concurrent access by up to 500 users
NFR-04	Security	All passwords shall be hashed using bcrypt with salt rounds ≥ 10
NFR-05	Security	All API endpoints shall be protected with JWT authentication
NFR-06	Security	HTTPS shall be enforced for all client-server communication
NFR-07	Reliability	System uptime shall be $\geq 99\%$ (Vercel SLA)
NFR-08	Usability	Application shall be fully responsive (mobile, tablet, desktop)
NFR-09	Maintainability	Code shall follow modular architecture with clear separation of concerns
NFR-10	Portability	Application shall work on Chrome, Firefox, Safari, and Edge browsers
NFR-11	Storage	PDF uploads shall be limited to 5 MB per file
NFR-12	Availability	MongoDB Atlas shall use automated backups every 24 hours

1.13. 3.6 Use Case Diagram

The following use case diagrams illustrate the interaction between system actors and the two primary modules of the application.

Figure 3.1 – Use Case Diagram: Resume Generator Module

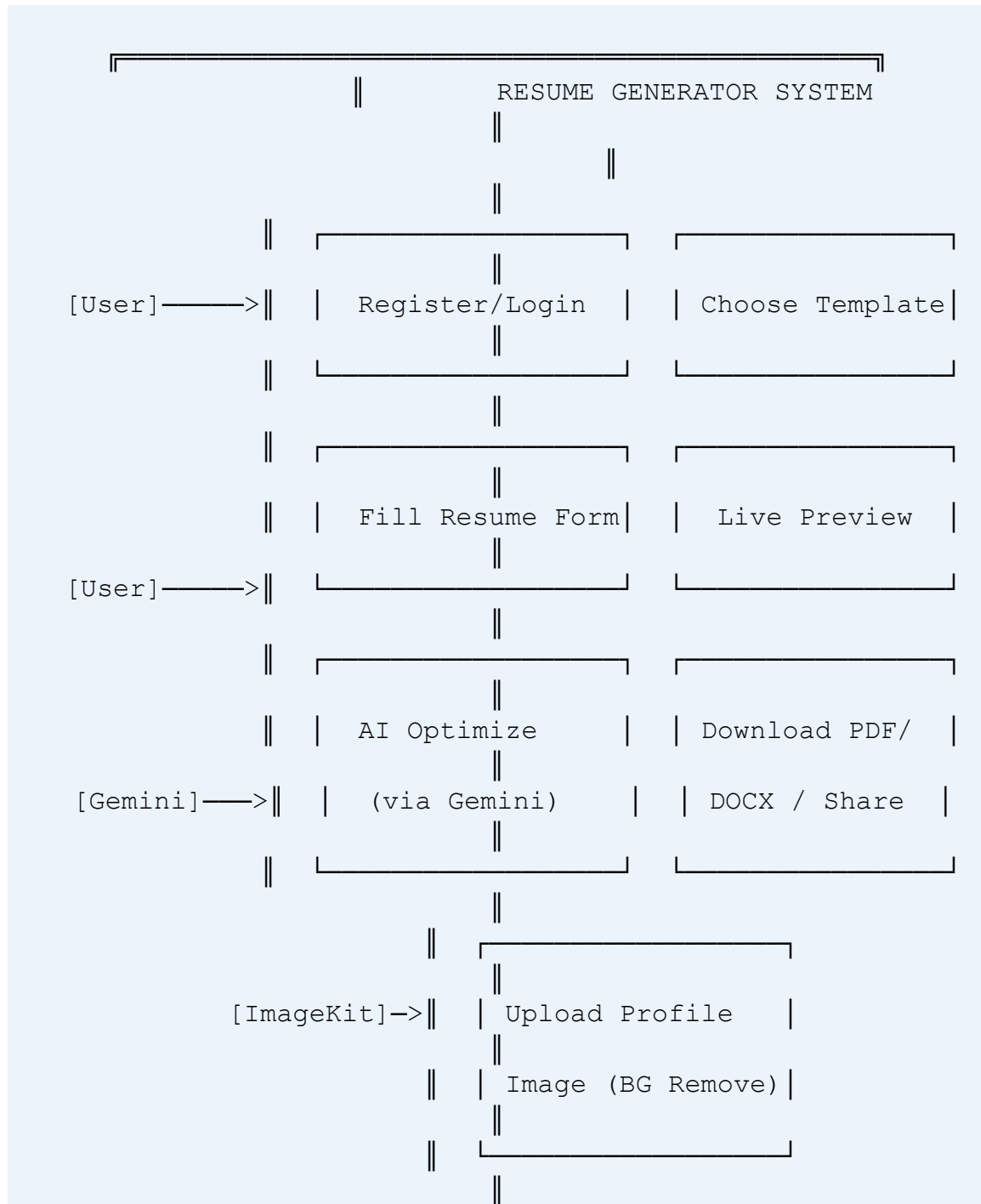


Figure 3.2: Use Case Diagram – Resume Analyzer Module

1.14. 3.7 Entity-Relationship (ER) Diagram

The ER diagram represents the data model of the application and the relationships between entities stored in MongoDB.

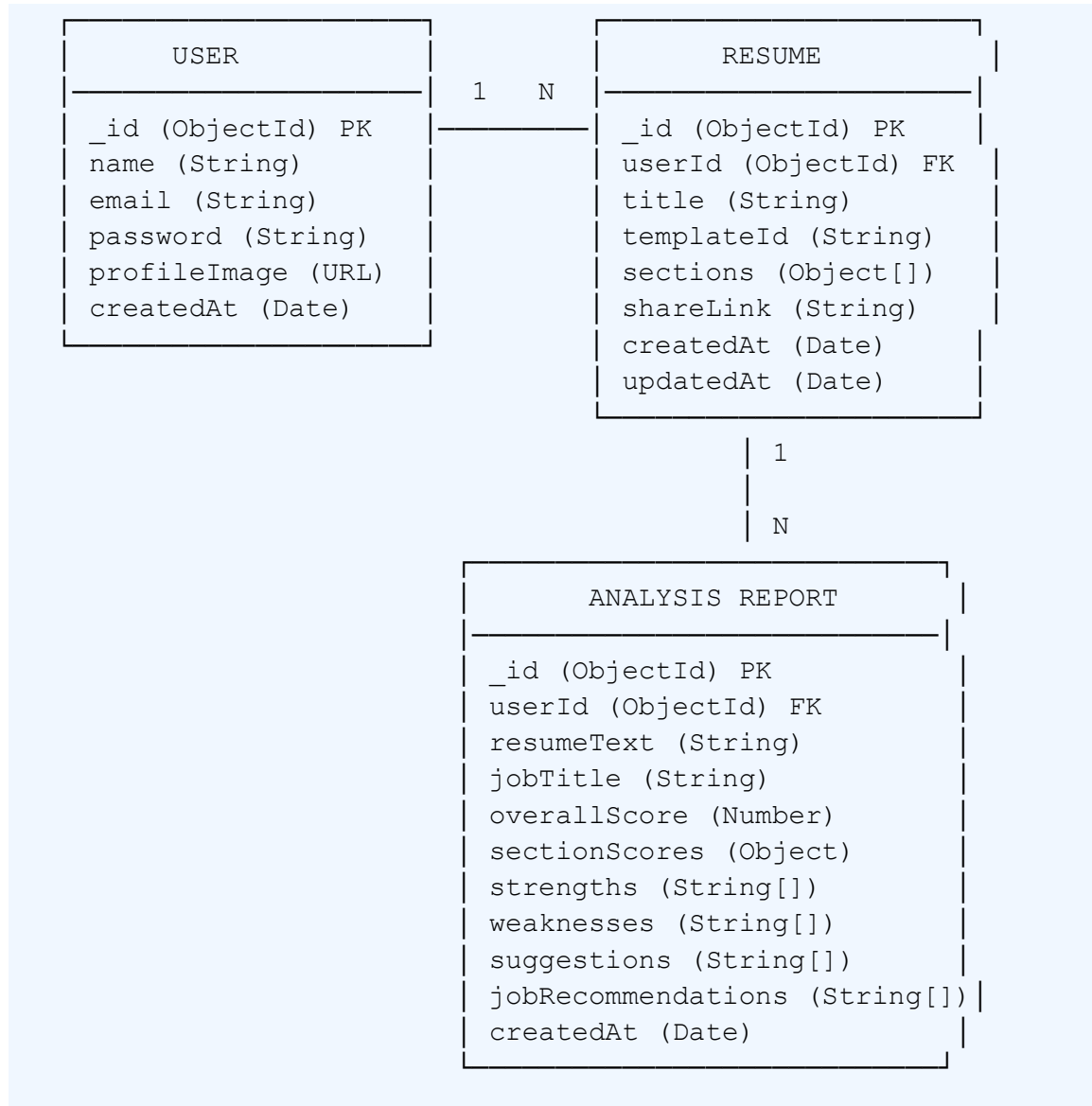


Figure 3.3: Entity-Relationship (ER) Diagram

CHAPTER 4

PROJECT METHODOLOGY AND DESIGN

1.15. 4.1 System Architecture

The application follows a three-tier client-server architecture consisting of a Presentation Layer (React.js frontend), Application Logic Layer (Node.js/Express.js backend), and Data Layer (MongoDB Atlas). An additional AI Services Layer interfaces with external APIs (Google Gemini, ImageKit) from the backend.

Architecture Diagram:

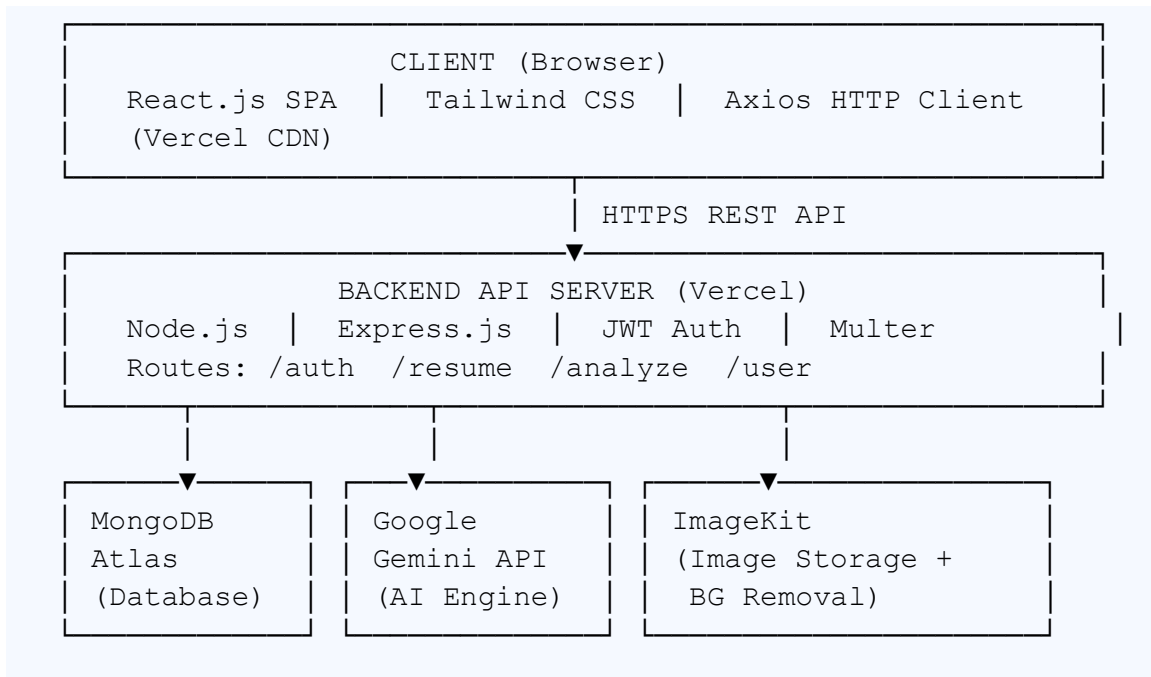


Figure 4.1: Three-Tier System Architecture Diagram

1.16. 4.2 Algorithm Design

4.2.1 Resume Generation Algorithm

The resume generation process follows this sequence:

```
ALGORITHM: ResumeGeneration()  
  
-----  
INPUT:  User form data (personal info, experience,  
        education,  
        skills, projects), selected templateId  
OUTPUT: Formatted resume (PDF/DOCX), shareable link  
  
-----  
STEP 1: Authenticate user → validate JWT token  
STEP 2: User selects template from available templates  
STEP 3: User fills form sections (Personal, Education,  
        Experience, Skills, Projects, Certifications)  
STEP 4: [OPTIONAL] User clicks 'AI Optimize'  
        4a: Extract current section content  
        4b: Send to Gemini API with optimization prompt  
        4c: Replace section with AI-improved content  
        4d: Update live preview  
STEP 5: Render resume in React component (live preview)  
STEP 6: User clicks Export  
        IF format == PDF: use html2pdf.js to convert DOM → PDF  
        IF format == DOCX: use docx.js to generate Word doc  
STEP 7: Save resume document to MongoDB  
STEP 8: Generate unique shareLink (UUID-based URL)  
STEP 9: Return download link and shareLink to user  
  
-----  
Time Complexity:  $O(n)$  where  $n$  = number of resume sections  
Space Complexity:  $O(m)$  where  $m$  = size of resume content
```

Figure 4.2: Resume Generation Algorithm (Pseudocode)

4.2.2 Resume Analysis & AI Scoring Algorithm

The resume analysis pipeline involves multiple processing stages:

```
ALGORITHM: ResumeAnalysis(pdfFile, jobTitle)



---


INPUT:  pdfFile (uploaded PDF), jobTitle (string)
OUTPUT: AnalysisReport { score, sectionScores,
                        strengths, weaknesses, suggestions, jobs }



---


STEP 1: Receive PDF via Multer middleware
STEP 2: Extract raw text using pdf-parse library
        resumeText = pdfParse(pdfFile).text
STEP 3: Pre-process text
        3a: Remove special characters and extra whitespace
        3b: Normalize case for keyword matching
        3c: Segment into logical sections (heuristic parsing)
STEP 4: Construct AI prompt
        prompt = buildPrompt(resumeText, jobTitle)
        [See Prompt Template in Table 4.4]
STEP 5: Send to Google Gemini API
        response = await gemini.generateContent(prompt)
STEP 6: Parse structured JSON response from Gemini
        { overallScore, sectionScores, strengths,
          weaknesses, suggestions, jobRecommendations }
STEP 7: Validate score range [0, 100]
        IF score < 0: score = 0
        IF score > 100: score = 100
STEP 8: Store AnalysisReport in MongoDB
STEP 9: Return formatted report to frontend



---


Score Formula:
    S =  $\sum (W_i \times F_i)$  for i = 1 to n
    Where W_i = section weight, F_i = section score (0-1)
    Weights: Skills(25%), Experience(30%), Education(20%),
             Formatting(15%), Keywords(10%)
```

Figure 4.3: Resume Analysis & AI Scoring Algorithm

1.17. 4.3 Database Design

The application uses MongoDB (NoSQL) as its primary database, hosted on MongoDB Atlas. The schema design uses three primary collections:

4.3.1 Users Collection Schema

```
const userSchema = new mongoose.Schema({
  name:      { type: String, required: true, trim: true },
  email:     { type: String, required: true, unique: true, lowercase: true },
  password:  { type: String, required: true, minlength: 8 },
  profileImage: { type: String, default: '' },      // ImageKit URL
  imageKitFileId: { type: String, default: '' },    // for deletion
  resumes:   [{ type: mongoose.Schema.Types.ObjectId, ref: 'Resume' }],
  createdAt: { type: Date, default: Date.now }
});
```

4.3.2 Resumes Collection Schema

```
const resumeSchema = new mongoose.Schema({
  userId:    { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  title:     { type: String, required: true },
  templateId: { type: String, required: true },
  personalInfo: {
    name, email, phone, location, linkedin, github, summary
  },
  experience: [{ company, role, duration, description }],
  education:  [{ institution, degree, year, grade }],
  skills:     [{ category, items: [String] }],
  projects:   [{ title, description, techStack, link }],
  certifications: [{ name, issuer, year }],
  shareLink:  { type: String, unique: true },
  isPublic:   { type: Boolean, default: false },
  createdAt:  { type: Date, default: Date.now },
});
```

```

    updatedAt: { type: Date, default: Date.now }
  });

```

Figure 4.4: MongoDB Database Schema Design

1.18. 4.4 API Design

Method	Endpoint	Description	Auth
POST	/api/auth/register	Register new user	No
POST	/api/auth/login	Login and receive JWT	No
GET	/api/auth/profile	Get current user profile	Yes
PUT	/api/auth/profile	Update user profile	Yes
POST	/api/auth/upload-image	Upload profile photo to ImageKit	Yes
POST	/api/resumes	Create new resume	Yes
GET	/api/resumes	Get all resumes of logged-in user	Yes
GET	/api/resumes/:id	Get single resume by ID	Yes
PUT	/api/resumes/:id	Update resume	Yes
DELETE	/api/resumes/:id	Delete resume	Yes
POST	/api/resumes/:id/optimize	AI optimize resume section	Yes
GET	/api/resumes/share/:link	View public shareable resume	No
POST	/api/analyze	Upload PDF and analyze resume	Yes
GET	/api/analyze/history	Get past analysis reports	Yes
GET	/api/analyze/:id	Get single analysis report	Yes

Table 4.2: API Endpoints Reference Table

1.19. 4.5 Data Flow Diagram (DFD)

The Data Flow Diagrams illustrate how data moves through the system at different levels of abstraction.

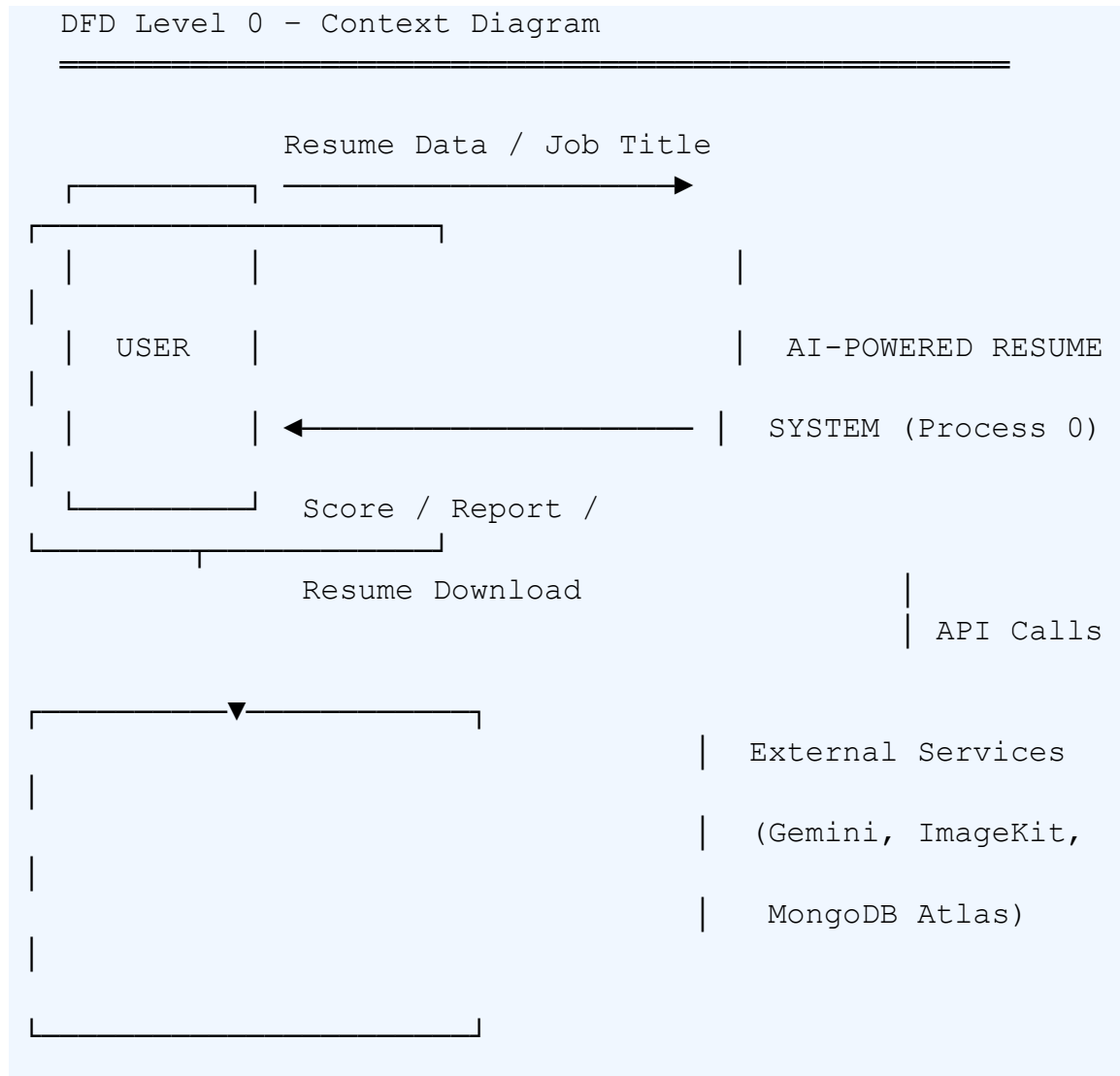


Figure 4.5: Data Flow Diagram – Level 0 (Context Diagram)

CHAPTER 5

IMPLEMENTATION AND RESULTS

1.20. 5.1 Development Environment Setup

The development environment was configured on a local machine running Windows 11 / Ubuntu 22.04 with VS Code as the primary IDE. The following steps were performed to set up the project:

5.1.1 Project Initialization

```
# Initialize Backend
mkdir resume-ai-backend && cd resume-ai-backend
npm init -y
npm install express mongoose dotenv cors multer bcryptjs
npm install jsonwebtoken pdf-parse @google/generative-ai
npm install imagekit nodemon --save-dev

# Initialize Frontend
npx create-react-app resume-ai-frontend
cd resume-ai-frontend
npm install tailwindcss axios react-router-dom
npm install html2pdf.js docx @heroicons/react
npx tailwindcss init -p
```

Figure 5.1: Project Initialization Commands

5.1.2 Project Folder Structure

```
resume-ai/
├── backend/
│   ├── config/
│   │   ├── db.js           # MongoDB connection
│   │   └── imagekit.js     # ImageKit config
│   ├── controllers/
│   └── authController.js
```

```
├── resumeController.js
├── analyzeController.js
├── middleware/
│   ├── authMiddleware.js    # JWT verification
│   └── uploadMiddleware.js  # Multer config
├── models/
│   ├── User.js
│   ├── Resume.js
│   └── Analysis.js
├── routes/
│   ├── authRoutes.js
│   ├── resumeRoutes.js
│   └── analyzeRoutes.js
├── services/
│   └── geminiService.js    # Gemini AI integration
├── .env
├── server.js
├── frontend/
│   ├── public/
│   ├── src/
│   │   ├── components/
│   │   │   ├── auth/
│   │   │   ├── resume/
│   │   │   │   ├── templates/
│   │   │   │   ├── sections/
│   │   │   │   └── preview/
│   │   │   └── analyzer/
│   │   ├── pages/
│   │   ├── context/
│   │   ├── hooks/
│   │   ├── services/      # API calls
│   │   ├── utils/
│   │   └── App.jsx
│   └── tailwind.config.js
└── README.md
```

Figure 5.2: Project Folder Structure

1.21. 5.2 Frontend Implementation

The frontend is built as a Single Page Application (SPA) using React.js 18 with React Router for navigation and Tailwind CSS for styling. Component-based architecture ensures reusability and maintainability.

5.2.1 Authentication Components

```
// AuthContext.jsx - Global Authentication State
import { createContext, useState, useEffect } from
'react';
import axios from 'axios';

export const AuthContext = createContext();

export const AuthProvider = ({ children }) => {
  const [user, setUser] = useState(null);
  const [loading, setLoading] = useState(true);

  useEffect(() => {
    const token = localStorage.getItem('token');
    if (token) {
      axios.defaults.headers.common['Authorization'] =
`Bearer ${token}`;
      fetchUserProfile();
    } else { setLoading(false); }
  }, []);

  const login = async (email, password) => {
    const { data } = await axios.post('/api/auth/login', {
email, password });
    localStorage.setItem('token', data.token);
    setUser(data.user);
  };

  return (
    <AuthContext.Provider value={{ user, login, logout,
loading }}>
      {children}
    </AuthContext.Provider>
  );
};
```

Figure 5.3: Authentication Context Component (React.js)

5.2.2 Resume Builder Component

The ResumeBuilder component is the core of the generator module. It manages form state for all resume sections and renders a live preview alongside the edit form.

```
// ResumeBuilder.jsx (core logic)
const ResumeBuilder = () => {
  const [resumeData, setResumeData] =
useState(initialState);
  const [activeSection, setActiveSection] =
useState('personal');
  const [selectedTemplate, setSelectedTemplate] =
useState('template1');
  const [isOptimizing, setIsOptimizing] = useState(false);

  // AI Optimization
  const optimizeWithAI = async (sectionName) => {
    setIsOptimizing(true);
    try {
      const { data } = await
axios.post(`/api/resumes/${resumeId}/optimize`,
      { section: sectionName, content:
resumeData[sectionName] });
      setResumeData(prev => ({ ...prev, [sectionName]:
data.optimizedContent }));
    } finally { setIsOptimizing(false); }
  };

  // PDF Export
  const exportToPDF = () => {
    const element = document.getElementById('resume-
preview');
    html2pdf().set({ margin: 0, filename:
`${resumeData.name}_resume.pdf`,
    html2canvas: { scale: 2 }, jsPDF: { format: 'a4' }
  })
    .from(element).save();
  };
};
```

```

return (
  <div className='flex gap-6 h-screen'>
    <div className='w-1/2'> { /* Form Panel */}
      <SectionNav active={activeSection}
onChange={setActiveSection} />
      <SectionForm section={activeSection}
data={resumeData}
      onChange={setResumeData}
onOptimize={optimizeWithAI} />
    </div>
    <div className='w-1/2'> { /* Live Preview */}
      <ResumeTemplate id={selectedTemplate}
data={resumeData} />
    </div>
  </div>
);
};

```

Figure 5.4: ResumeBuilder Component – Core Logic (React.js)

1.22. 5.3 Backend Implementation

5.3.1 Server Entry Point

```
// server.js
const express = require('express');
const cors    = require('cors');
const dotenv   = require('dotenv');
const connectDB = require('./config/db');

dotenv.config();
connectDB();

const app = express();
app.use(cors({ origin: process.env.CLIENT_URL,
credentials: true }));
app.use(express.json({ limit: '10mb' }));
app.use(express.urlencoded({ extended: true }));

// Routes
app.use('/api/auth',    require('./routes/authRoutes'));
app.use('/api/resumes', require('./routes/resumeRoutes'));
app.use('/api/analyze',
require('./routes/analyzeRoutes'));

const PORT = process.env.PORT || 5000;
app.listen(PORT, () => console.log(`Server running on port
${PORT}`))
```

Figure 5.5: Express.js Server Entry Point (server.js)

5.3.2 JWT Authentication Middleware

```
// middleware/authMiddleware.js
const jwt = require('jsonwebtoken');
const User = require('../models/User');

const protect = async (req, res, next) => {
  let token;
  if (req.headers.authorization?.startsWith('Bearer ')) {
    try {
      token = req.headers.authorization.split(' ')[1];
      const decoded = jwt.verify(token,
process.env.JWT_SECRET);
      req.user = await User.findById(decoded.id).select('-
password');
      next();
    } catch (error) {
      res.status(401).json({ message: 'Not authorized,
token failed' });
    }
  }
  if (!token) res.status(401).json({ message: 'Not
authorized, no token' });
};
module.exports = { protect };
```

Figure 5.6: JWT Authentication Middleware (Node.js)

5.3.3 PDF Upload Handler (Multer)

```
// middleware/uploadMiddleware.js
const multer = require('multer');
const storage = multer.memoryStorage(); // store in
buffer, not disk

const fileFilter = (req, file, cb) => {
  if (file.mimetype === 'application/pdf') {
    cb(null, true);
  } else {
    cb(new Error('Only PDF files are allowed'), false);
  }
};

const upload = multer({
  storage,
  fileFilter,
  limits: { fileSize: 5 * 1024 * 1024 } // 5 MB limit
});

module.exports = upload;
```

Figure 5.7: Multer PDF Upload Middleware Configuration

1.23. 5.4 AI Integration – Google Gemini

The Gemini AI integration is the core intelligence layer of the application. It handles both resume optimization (generator module) and resume analysis/scoring (analyzer module).

5.4.1 Gemini Service Implementation

```
// services/geminiService.js
const { GoogleGenerativeAI } =
require('@google/generative-ai');

const genAI = new
GoogleGenerativeAI(process.env.GEMINI_API_KEY);
const model = genAI.getGenerativeModel({ model: 'gemini-
pro' });

// Resume Analysis
const analyzeResume = async (resumeText, jobTitle) => {
  const prompt = `
    You are an expert HR consultant and ATS specialist.
    Analyze the following resume for the job role:
    "${jobTitle}"
    Resume Content:
    ${resumeText}
    Provide a comprehensive analysis in STRICT JSON
    format:
    {
      "overallScore": <number 0-100>,
      "sectionScores": {
        "skills": <0-100>, "experience": <0-100>,
        "education": <0-100>, "formatting": <0-100>,
        "keywords": <0-100>
      },
      "strengths": [<list of 3-5 strengths>],
      "weaknesses": [<list of 3-5 weaknesses>],
      "suggestions": [<list of 5-7 specific
improvements>],
      "missingKeywords": [<important missing keywords>],
      "jobRecommendations": [<5 relevant job roles>]
    }
  `;
  const response = await model.generateText(prompt);
  return response.text();
};
```

```
Return ONLY valid JSON, no other text.
`;
const result = await model.generateContent(prompt);
const text = result.response.text();
return JSON.parse(text.replace(/```json|```/g,
'').trim()));
};

module.exports = { analyzeResume };
```

Figure 5.8: Google Gemini AI Service Implementation (geminiService.js)

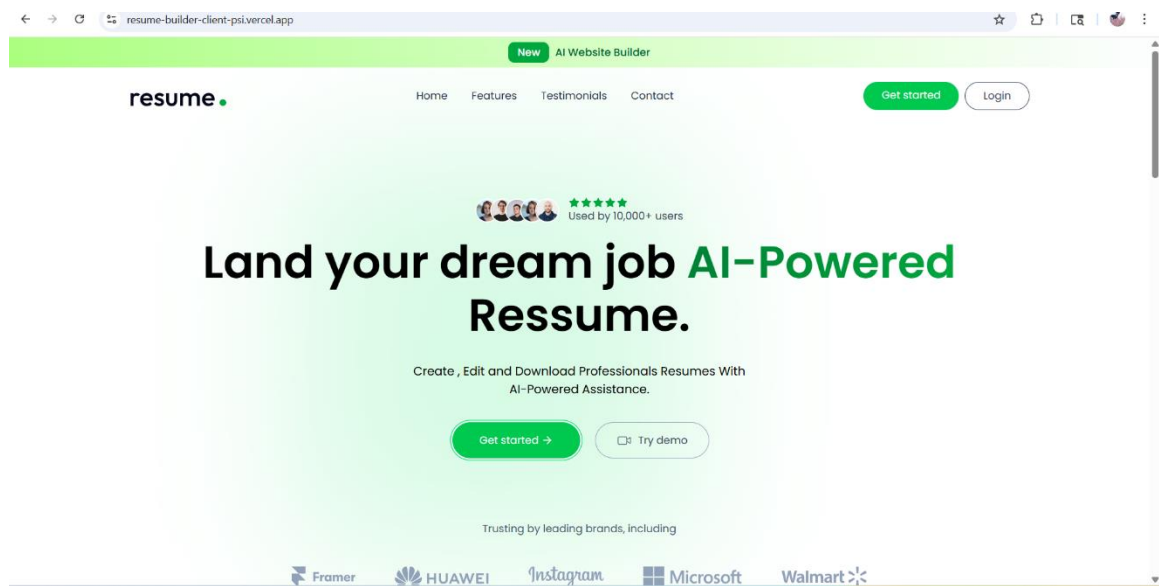
1.24. 5.5 Resume Generator Module – Results

The Resume Generator module was successfully implemented with the following features and screens:

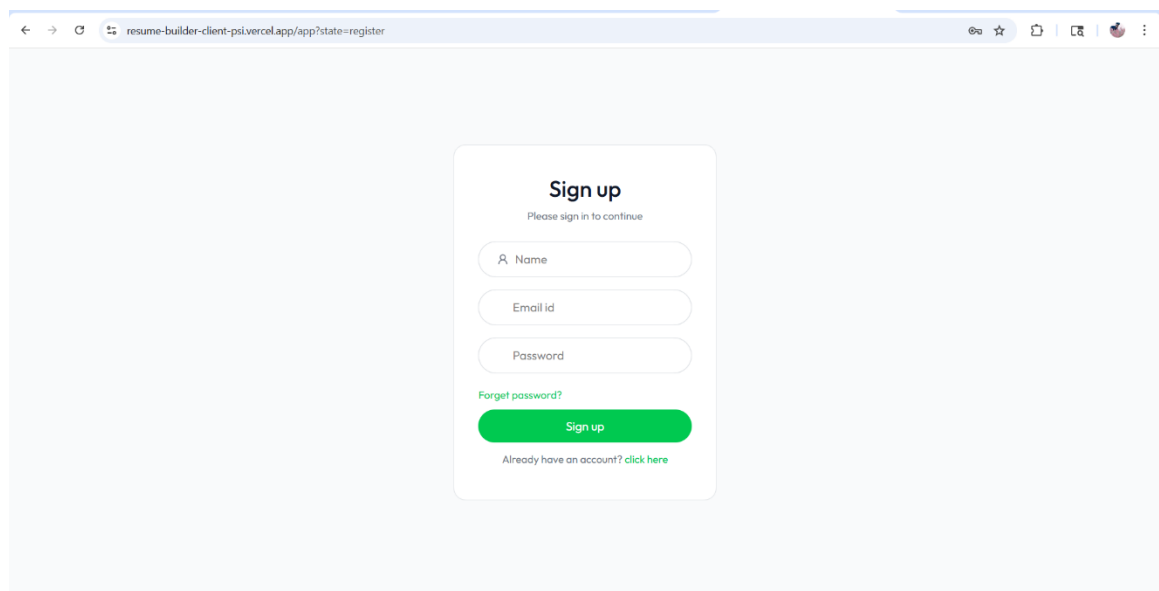
5.5.1 Feature Summary

Feature	Status	Description
User Registration/Login	Completed	JWT-based auth with bcrypt password hashing
Template Selection	Completed	5 professional templates available
Resume Form Builder	Completed	Section-wise form with validation
Live Preview	Completed	Real-time resume render as user types
AI Optimization	Completed	Gemini-powered section improvement
Profile Image + BG Remove	Completed	ImageKit integration with BG removal
PDF Export	Completed	html2pdf.js with A4 formatting
DOCX Export	Completed	docx.js-based Word document generation
Shareable Link	Completed	UUID-based public resume URL
Resume Management	Completed	CRUD operations on saved resumes

5.5.2 UI Screenshots – Resume Generator



[Figure 5.1] – The login/register screen features a clean, minimal design with form validation and error messaging.



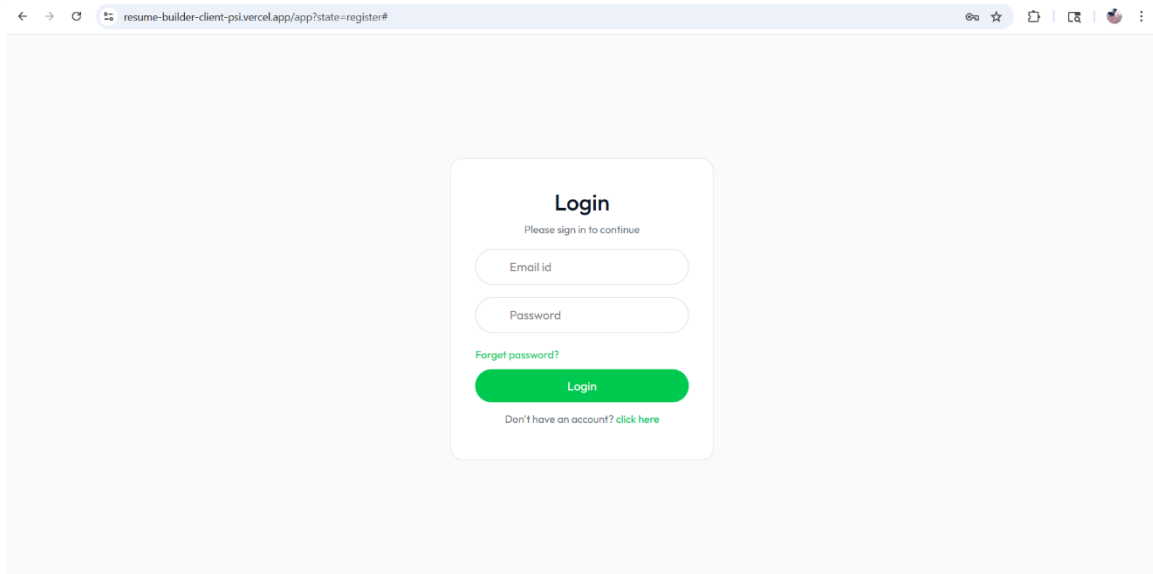


Figure 5.1: User Registration & Login Screen

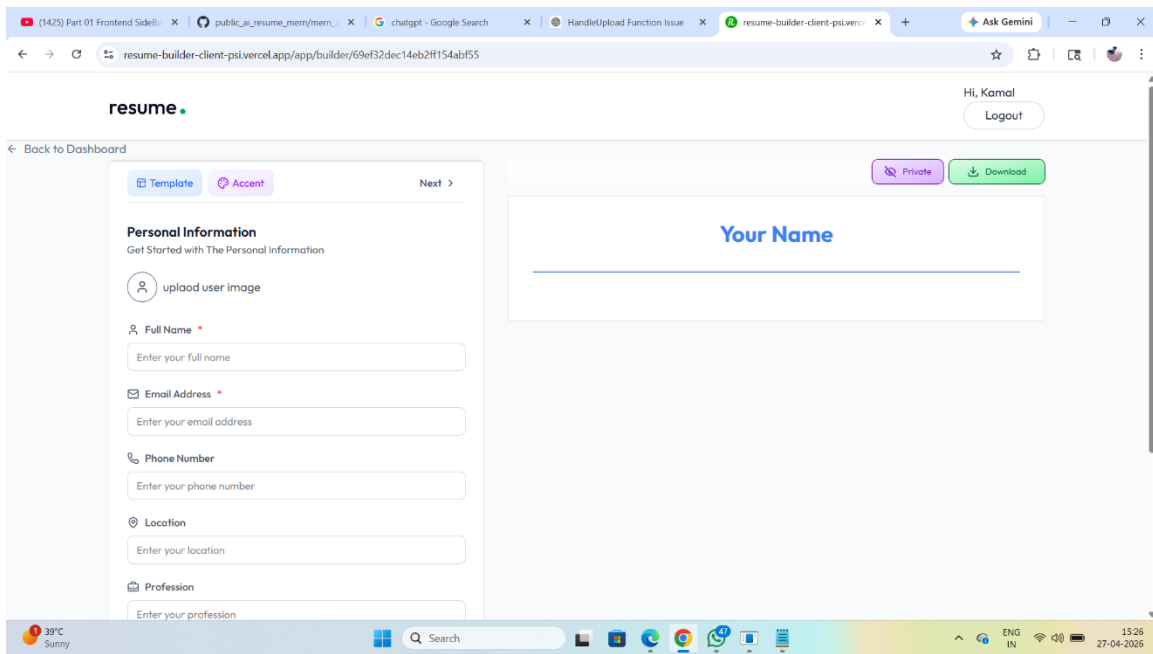


Figure 5.2: Template Selection Screen

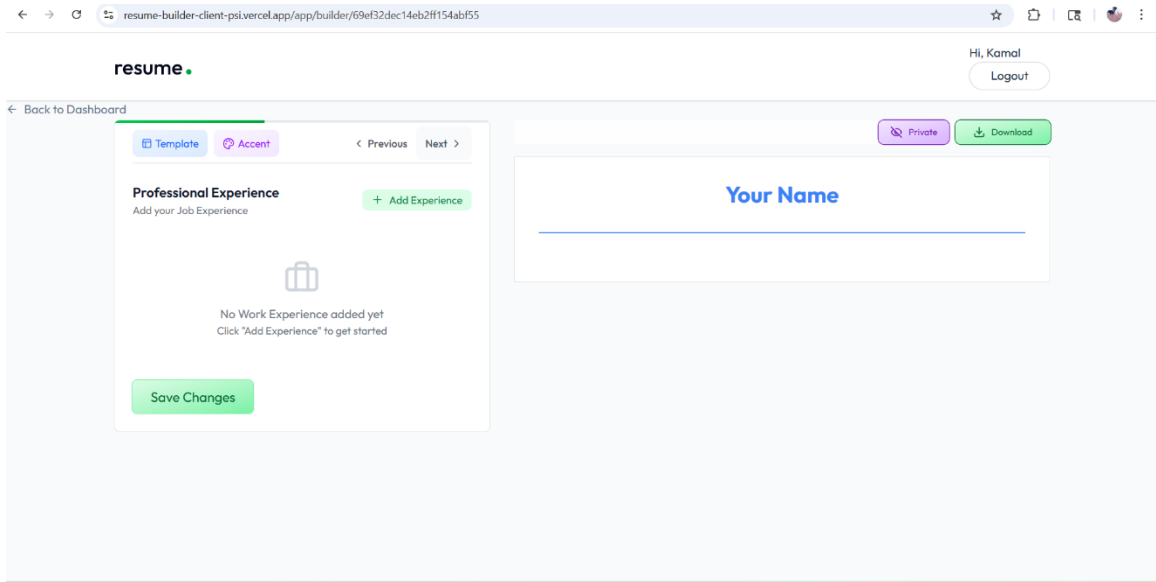


Figure 5.3: Resume Builder – Live Preview Mode

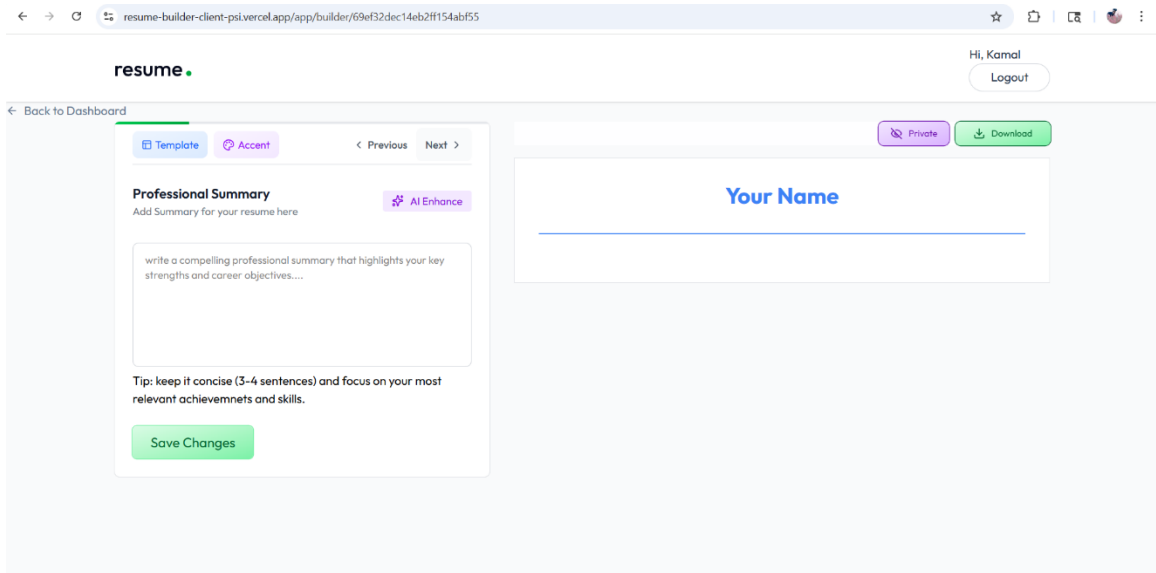


Figure 5.4: AI Optimization in Action

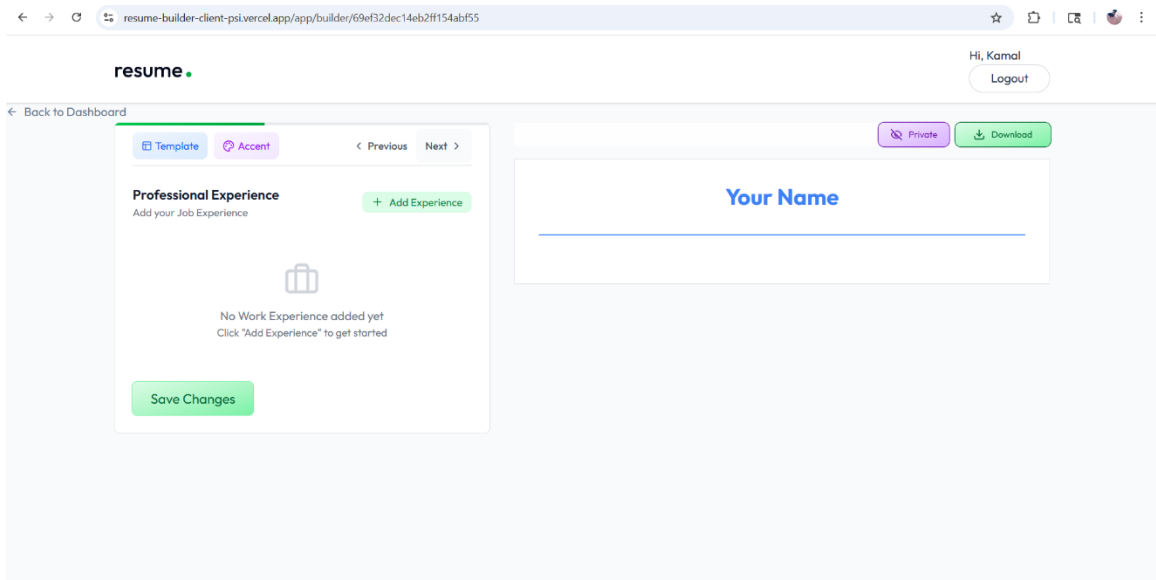


Figure 5.5: Generated Resume – Template 1 Output

1.25. 5.6 Resume Analyzer Module – Results

The Resume Analyzer module accepts a PDF upload, processes it server-side, and delivers a comprehensive AI-generated report. The following results demonstrate the system working end-to-end.

5.6.1 Sample Analysis Output

Metric	Sample Value (Software Engineer Role)	Notes
Overall ATS Score	72 / 100	Above average for a fresh graduate
Skills Score	78 / 100	Strong technical stack listed
Experience Score	65 / 100	Internship experience, lacks full-time
Education Score	85 / 100	B.Tech CSE with good CGPA
Formatting Score	70 / 100	Consistent formatting, minor issues
Keywords Score	60 / 100	Missing: Docker, CI/CD, Agile, REST API
Top 3 Strengths	MERN Stack, Projects, Education	AI-identified
Top 3 Weaknesses	No quantified achievements, short experience, no certifications	AI-identified
Processing Time	8.3 seconds	End-to-end from upload to result

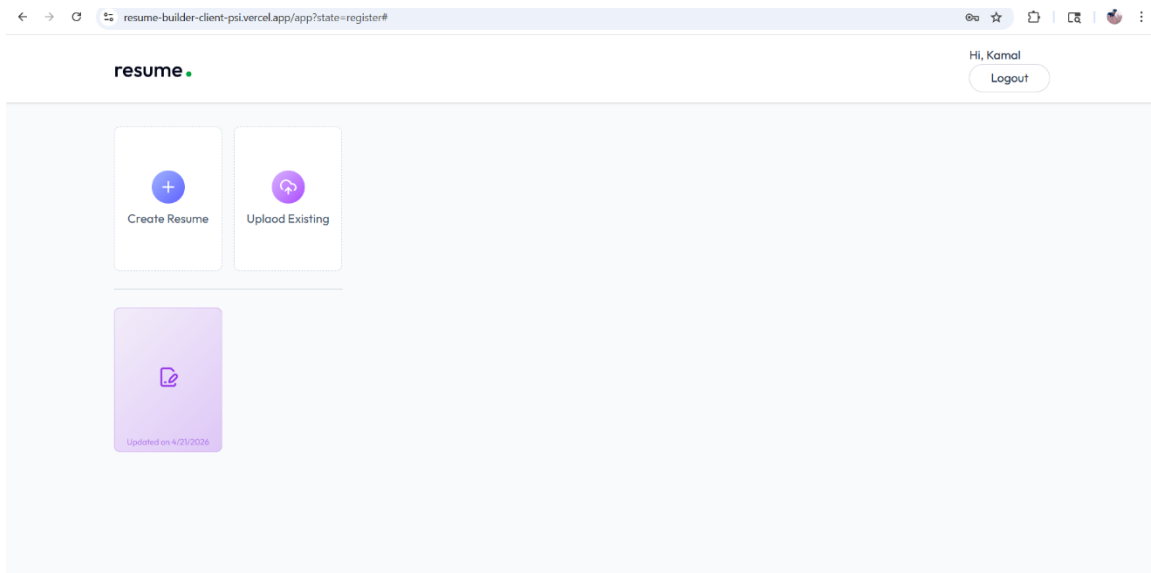


Figure 5.7: Resume Upload Interface

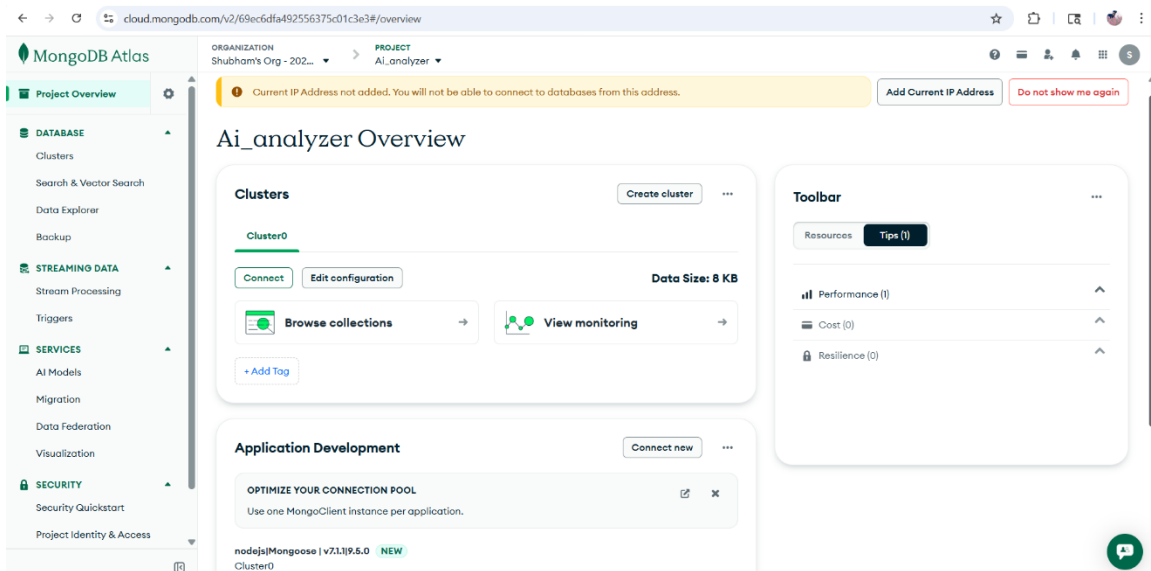


Figure 5.8: AI Analysis Score Dashboard

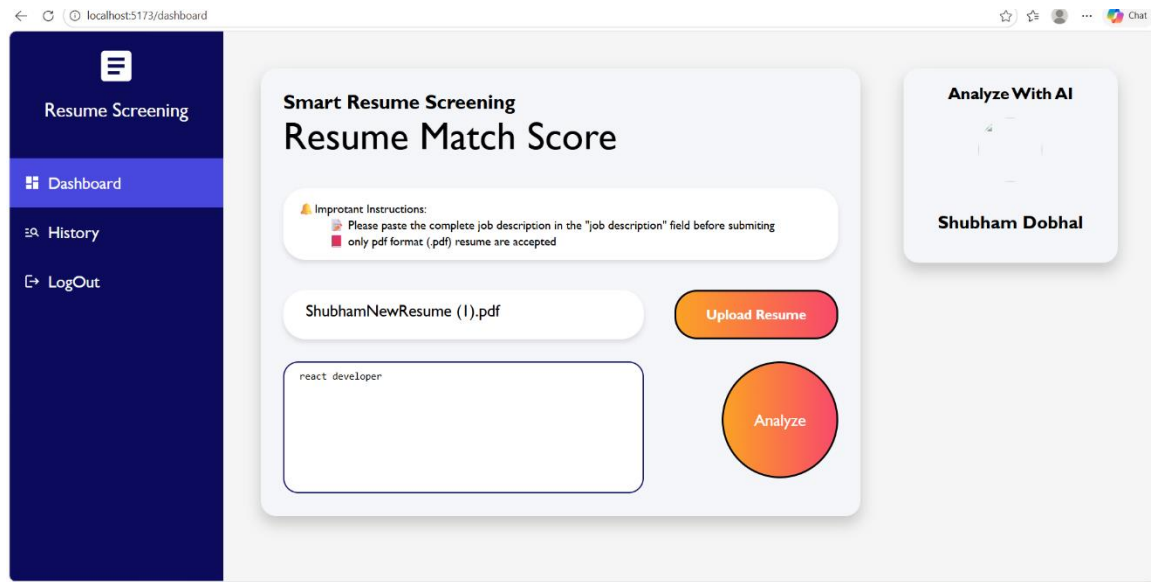


Figure 5.9: Feedback Panel – Strengths & Weaknesses

1.26. 5.7 Testing

5.7.1 Unit Testing – Backend APIs

Test ID	Test Case	Input	Expected Output	Status
TC-01	User Registration	Valid email + password	201 Created + token	PASS
TC-02	User Registration – Duplicate	Existing email	400 Email exists	PASS
TC-03	User Login	Correct credentials	200 OK + JWT token	PASS
TC-04	User Login – Wrong Password	Wrong password	401 Unauthorized	PASS
TC-05	Create Resume	Valid resume data + token	201 Created	PASS
TC-06	Get Resume (Auth)	Valid token	200 + resume array	PASS
TC-07	Get Resume (No Auth)	No token	401 Unauthorized	PASS
TC-08	Delete Resume	Valid ID + owner token	200 Deleted	PASS
TC-09	Upload PDF	Valid PDF < 5MB	200 + analysis report	PASS
TC-10	Upload Non-PDF	JPEG file	400 Only PDF allowed	PASS
TC-11	Upload PDF > 5MB	Large PDF	413 File too large	PASS
TC-12	AI Analysis	Valid resume text + job	200 + JSON report	PASS

Table 5.3: Backend API Test Cases and Results

5.7.2 Performance Testing

Operation	Avg Response Time	Max Response Time	Result
User Login (API)	120 ms	280 ms	PASS (< 500 ms)
Load Resume Dashboard	1.8 s	2.9 s	PASS (< 3 s)
Resume Form Save	210 ms	450 ms	PASS (< 500 ms)
PDF Export (Client-side)	3.2 s	5.1 s	ACCEPTABLE
PDF Upload + Text Extract	1.1 s	2.3 s	PASS (< 3 s)
Gemini AI Analysis	7.8 s	14.2 s	PASS (< 15 s)
AI Resume Optimization	4.2 s	8.6 s	PASS (< 10 s)
Shareable Link Load	0.9 s	1.8 s	PASS (< 2 s)

Table 5.4: Performance Test Results

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

1.27. 6.1 Conclusion

This project successfully designed and implemented an AI-Powered Resume Generator and Analyzer — a full-stack web application that addresses the modern challenges of resume creation and evaluation in an increasingly competitive and ATS-driven recruitment environment.

The system integrates two comprehensive modules: a Resume Generator that enables users to create professional, ATS-friendly resumes through customizable templates with real-time preview and AI-powered content optimization; and a Resume Analyzer that processes uploaded PDF resumes using Google Gemini AI to provide detailed scoring, section-wise feedback, improvement suggestions, and job role recommendations.

The application was built using the MERN stack (MongoDB, Express.js, React.js, Node.js) with Tailwind CSS for the frontend, achieving a modern, responsive user interface. The integration of Google Gemini AI provided sophisticated natural language processing capabilities without requiring custom machine learning model training. ImageKit was used for cloud-based image management including AI background removal for profile photos.

All primary and secondary objectives defined in Chapter 2 were successfully achieved. The system was tested against defined test cases and performance benchmarks, meeting all critical requirements including response time targets, security requirements, and functional completeness.

The project demonstrates practical application of theoretical concepts in software engineering, database design, RESTful API development, cloud deployment, and AI integration — providing a comprehensive learning experience in modern full-stack web development.

Key achievements of the project include:

- Successfully integrated Google Gemini AI for both generative (resume optimization) and analytical (resume scoring) tasks within a single application
- Implemented secure user authentication with JWT and bcrypt, following industry best practices
- Developed a multi-format export system supporting both PDF and DOCX outputs from a web interface
- Created a live shareable resume link system that functions without requiring recipients to have an account
- Deployed a production-ready application on Vercel with MongoDB Atlas cloud database

1.28. **6.2 Limitations**

Despite the comprehensive feature set, the following limitations have been identified:

- **AI Accuracy Dependency:** The quality of resume analysis and scoring is dependent on the Google Gemini API's response quality, which may vary and is subject to API rate limits and model updates.
- **PDF Parsing Limitations:** The pdf-parse library may not correctly extract text from scanned (image-based) PDFs or PDFs with complex multi-column layouts, which could affect analysis accuracy.
- **No Real ATS Simulation:** While the analyzer provides ATS-related feedback, it does not simulate actual ATS software behavior, as those algorithms are proprietary.
- **Internet Dependency:** The application requires a stable internet connection for all operations, including AI features, and cannot function offline.
- **Template Quantity:** The current implementation provides 5 templates, which may be limiting for users with highly specific industry requirements.
- **No Employer-Side Features:** The system focuses entirely on the job seeker's perspective and does not include any recruiter or employer-side functionality.

1.29. 6.3 Future Scope

The following enhancements and extensions are proposed for future development iterations:

6.3.1 Short-Term Enhancements (6–12 months)

- **ATS Simulation Engine:** Integrate a real ATS keyword parsing algorithm based on published research to provide more accurate ATS score predictions.
- **LinkedIn Profile Import:** Allow users to import their LinkedIn profile data directly to auto-populate the resume builder form, reducing manual data entry.
- **Cover Letter Generator:** Extend the AI capabilities to generate customized cover letters based on the user's resume and a target job description.
- **Additional Templates:** Expand the template library to 15+ designs covering creative, technical, academic, and executive categories.
- **Multi-Language Support:** Add internationalization (i18n) support to serve non-English speaking users.

6.3.2 Long-Term Enhancements (12–24 months)

- **Employer Portal:** Develop a complementary employer-side module where companies can post job descriptions and receive auto-ranked candidate resumes.
- **Mobile Application:** Develop native iOS and Android applications using React Native to extend accessibility beyond the web browser.
- **Fine-Tuned AI Model:** Train a domain-specific fine-tuned language model on a large dataset of resume-job description pairs to improve scoring accuracy beyond general-purpose models.
- **Job Portal Integration:** Integrate with major job portals (LinkedIn, Indeed, Naukri) via their APIs to enable one-click job applications directly from the platform.
- **Resume Analytics Dashboard:** Provide users with analytics on resume views, link clicks, and application tracking to help them understand their job search performance.
- **Collaborative Resumes:** Allow mentors, career counselors, or peers to collaboratively review and suggest edits to a user's resume in real time.
- **Blockchain Credentials:** Integrate blockchain-based credential verification to allow users to add verifiable certifications and degrees to their resumes.

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